

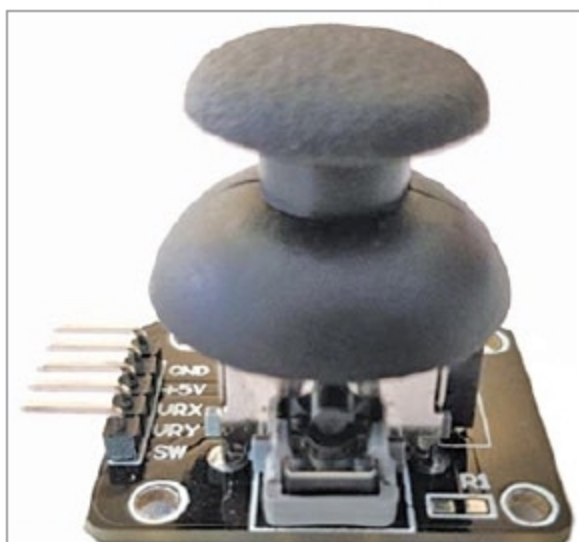


Fig. 3: Joystick module

former (X1). In this project, mainly two voltages are required: 12V and 5V. 12V DC is used for relays, and 5V for Arduino Nano MCU.

Arduino Nano. It is a small board compatible with ATmega328 MCU. It has the following features:

- Operating voltage (logic level): 5V
- Input voltage (recommended): 7V-12V
- Input voltage (limits): 6V-20V
- Fourteen digital I/O pins (of which six provide PWM output)

- Eight analogue input pins
- DC current per I/O pin: 40mA
- Flash memory: 32kB (ATmega 328), of which 2kB is used by boot-loader

- SRAM: 2kB (ATmega328)
- EEPROM: 1kB
- Clock speed: 16MHz

Operation

When the joystick module is in ideal state, digital value for both VRx and Vry is 512. ADC of Arduino is ten-bit, and digital value varies from 0 to 1023 levels according to analogue voltage received from the joystick. Ideally, the joystick is in midpoint or standby mode, so value shows is 512.

Case 1. When you move the shaft in x-left direction, digital value is less than 50. In this condition, pin D2 of Board1 is made high from the program, BC547 transistor T1 conducts, relay RL1 is energised and load connected across RL1 turns on. Diode D3 is used as freewheeling diode to eliminate back EMF.

Case 2. When you move the shaft in x-right direction, digital value is greater than 900. In this condition, pin D3 of Board1 is made high, BC547 transistor T2 conducts, relay RL2 is energised and electrical load/device connected across RL2 turns on.

Case 3. When you move the shaft in y-down direction, digital value is less than 50. Here, pin D4 is made high, BC547 transistor T3 conducts, relay RL3 is energised and load connected across RL3 turns on.

Case 4. When you move the shaft in y-up direction, digital value is greater than 900. Here, pin D5 of Board1 is

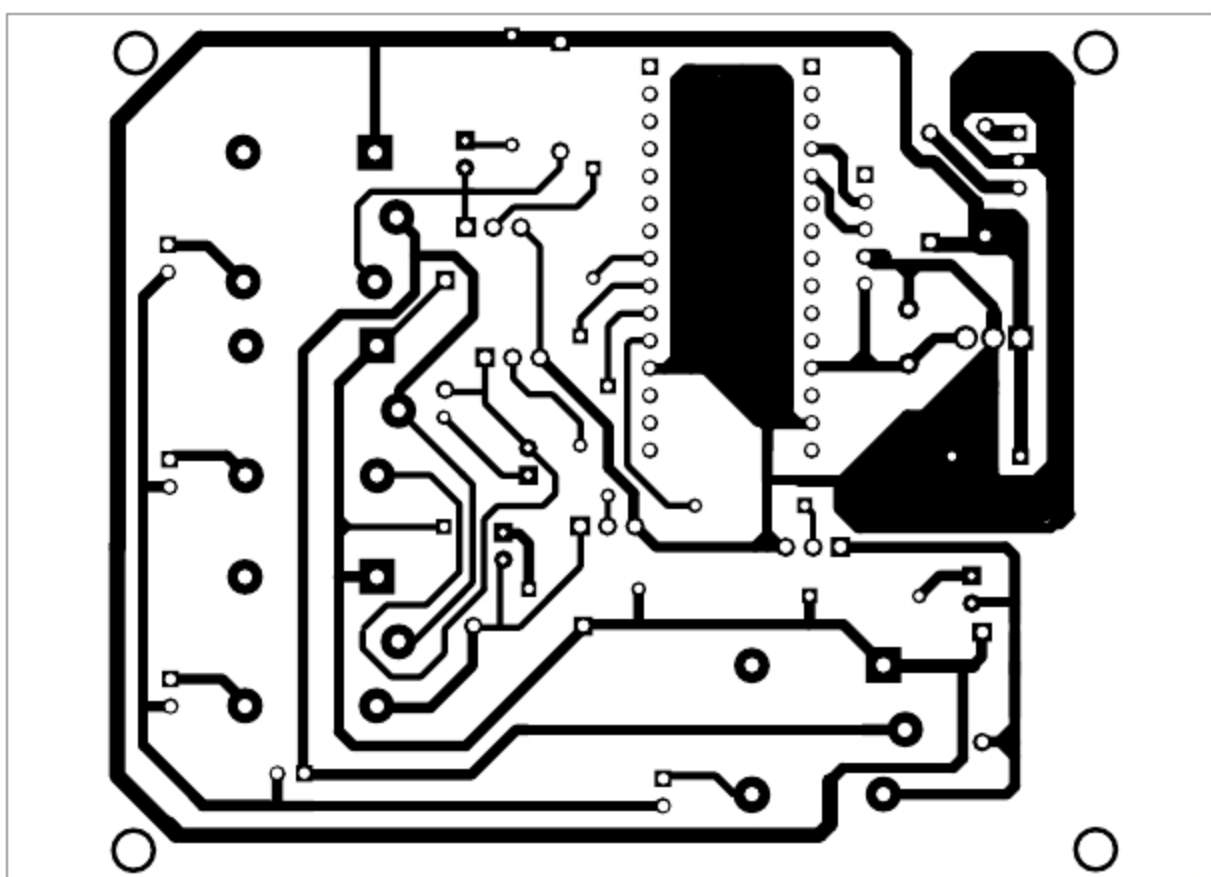


Fig. 4: Actual-size PCB layout of the joystick-controlled industrial automation system

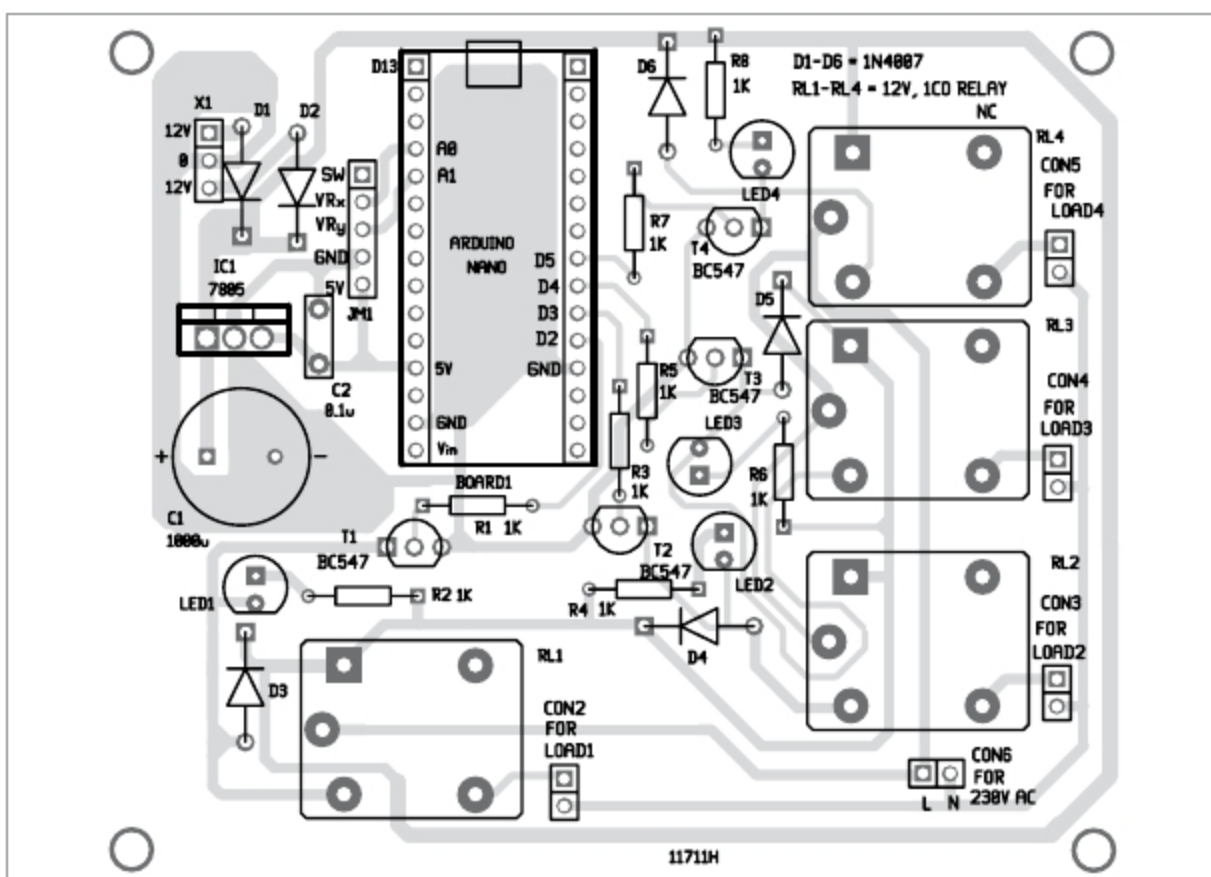


Fig. 5: Components layout for the PCB

PARTS LIST

Semiconductors:

- IC1 - 7805, 5V regulator
- Board1 - Arduino Nano board
- T1-T4 - BC547 npn transistor
- D1-D6 - 1N4007 rectifier diode
- LED1-LED4 - 5mm LED

Resistors (all 1/4-watt, $\pm 5\%$ carbon):

- R1-R8 - 1-kilo-ohm

Capacitors:

- C1 - 1000 μ F, 35V electrolytic
- C2 - 0.1 μ F ceramic

Miscellaneous:

- JM1 - 5-pin joystick module
- RL-RL4 - 12V, single-changeover relay
- X1 - 230V AC primary to 12V-0-12V, 750mA secondary transformer

(All these are available from kitsnspares.com)